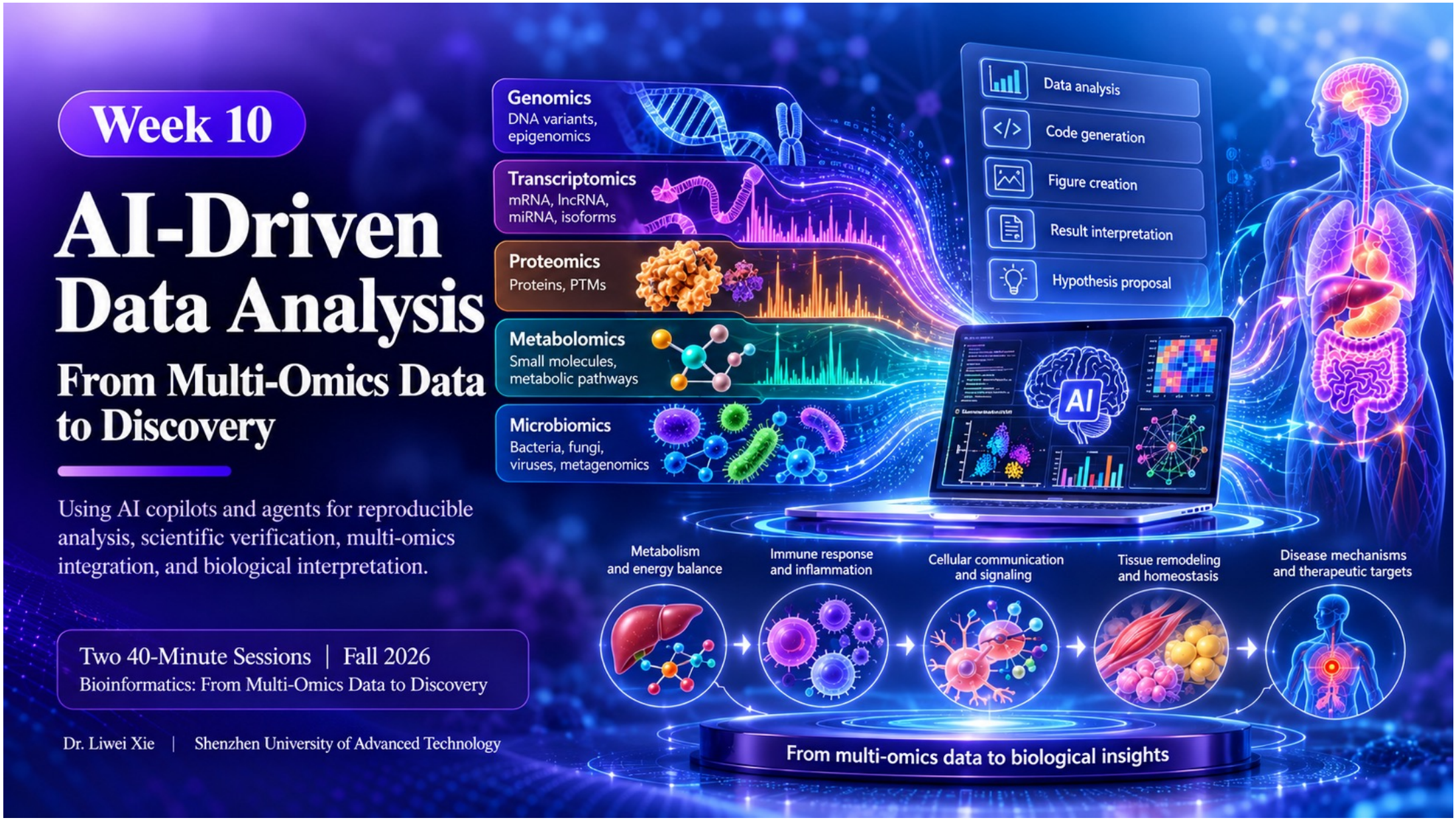

Week 10
AI-Driven Data Analysis
From Multi-Omics Data to Discovery
Using AI copilots and agents for reproducible analysis, scientific verification, multi-omics integration, and biological interpretation.
Two 40-Minute Sessions | Fall 2026
Bioinformatics: From Multi-Omics Data to Discovery
Dr. Liwei Xie | Shenzhen University of Advanced Technology
Genomics
DNA variants, epigenomics
Transcriptomics
mRNA, lncRNA, miRNA, isoforms
Proteomics
Proteins, PTMs
Metabolomics
Small molecules, metabolic pathways
Microbiomics
Bacteria, fungi, viruses, metagenomics
Data analysis
Code generation
Figure creation
Result interpretation
Hypothesis proposal
AI
Metabolism and energy balance
Immune response and inflammation
Cellular communication and signaling
Tissue remodeling and homeostasis
Disease mechanisms and therapeutic targets
From multi-omics data to biological insights

Data Visualization, Scientific Polishing & AI-Driven Data Analysis

From omics data to clear, reproducible, and biologically meaningful figures

1 Biological Samples and Multi-Omics Data

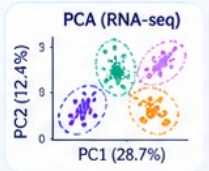


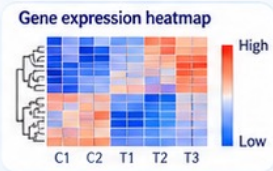
Human / Model Organisms

- Genomics
- Transcriptomics
- Microbiome
- Proteomics
- Metabolomics
- Single-cell
- Spatial maps

2 Data Processing & Analysis in R / ggplot2

- Quality control
- Normalization & transformation
- Statistical modeling (e.g. DE, FDR, CI)
- Reproducible code and pipelines





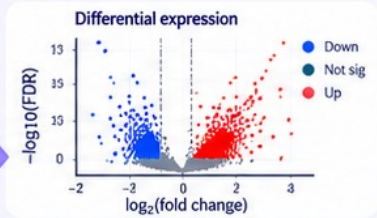
3 AI-Assisted Review and Enhancement

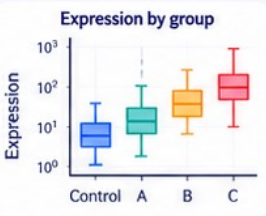
- Draft code & analyses
- Check assumptions
- Refactor plots
- Improve clarity
- Verify outputs

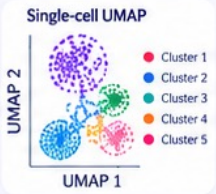
AI + Human Expertise

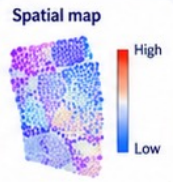
Code review · Plot optimization · Assumption checks · Biological interpretation

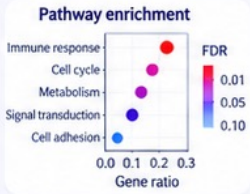
4 Polished Scientific Figures and Reproducible Outputs











1 Visual Grammar

Choose the right plot for the data and question.



Types · Axes · Scales · Layers

2 Scientific Polishing

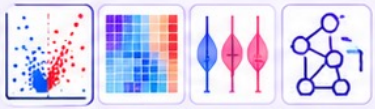
Improve hierarchy, labels, colors, and layout.



Colours · Labels · Consistent Style

3 Figure Storytelling

Assemble multi-panel figures with a clear narrative.



Multi-panel · Flow · Captions

4 AI-Driven Analysis

Use AI to draft code, audit graphics, and support interpretation.



Code · Review · Refine · Interpret

5 Independent Verification

Check data, code, and claims for reproducibility.

- Data sources
- Code & parameters
- Statistical assumptions
- Figure provenance



Reproducibility · Transparency

Good figures do not just look clear—they preserve the biological question, the statistical object, and the evidence.

From Omics Data to Scientific Evidence

Data → R code → visual evidence → AI review

1 Biological Question and Study Design



Human / Model Organisms

- Genomics
- Transcriptomics
- Microbiome
- Proteomics
- Metabolomics
- Single-cell
- Spatial maps

Define the question

- Biological hypothesis
- Experimental design
- Samples and groups
- Unit of replication

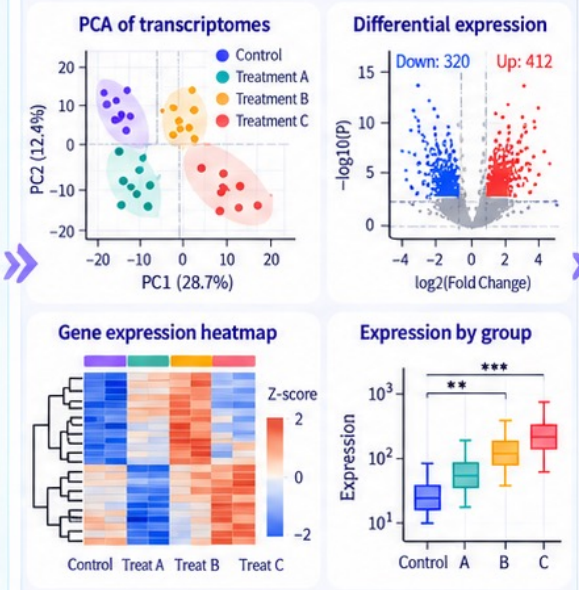
2 Data Processing and Analysis in R

- Quality control
- Normalization & transformation
- Statistical modeling (e.g. DE, FDR)
- Reproducible code and pipelines

```
library(ggplot2)
ggplot(data, aes(...)) +
  geom_point() +
  theme_bw()
```

Data in | Processing | R scripts | Reproducible pipeline

3 Visual Evidence from the Data



4 AI-Assisted Review and Verification

- Check figure accuracy
- Verify statistical claims
- Assess biological plausibility
- Suggest improvements
- Ensure reproducibility

AI + Human Expertise

- Figure review
- Assumption checks
- Code review
- Interpretation guidance

5 Scientific Interpretation

- Clear and honest figures
- Biologically meaningful conclusions
- Reproducible analysis
- Transparent methods
- Ready to share and publish

From figures to evidence

- Answer the biological question
- Support with statistical evidence
- Reproducible artifact (code + figure + notes)

Biological question | Unit of replication | Reproducible visual grammar

A defensible figure begins with study design and ends with an audited, reproducible artifact.

Can a Beautiful Figure Be Scientifically Wrong?

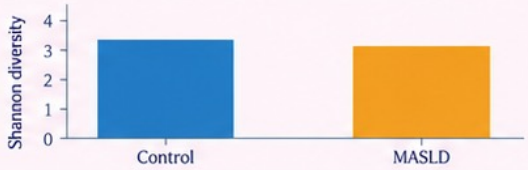
Visual appeal cannot rescue a flawed statistical object.

Common pitfalls

Pretty figures that hide or distort the truth

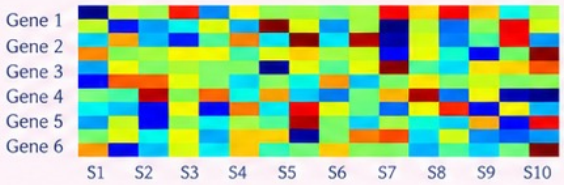
1 Means only (hides variability)

Bar heights show only means and conceal the distribution and uncertainty.



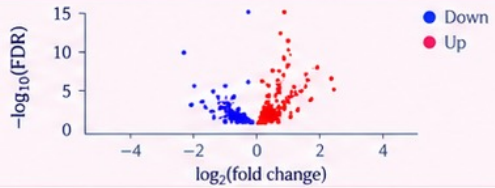
2 Rainbow heatmap (arbitrary scale)

Rainbow color schemes can create artificial boundaries and obscure true patterns.



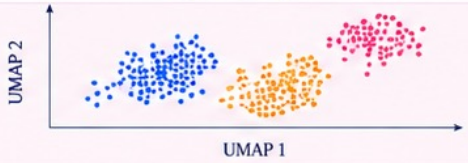
3 Incomplete volcano (missing non-significant)

Showing only significant points exaggerates effects and omits context.



4 Overclaimed embedding

Unlabeled clusters and no biological context can mislead interpretation.



Better scientific repairs

Show the data, respect the scale, and be transparent

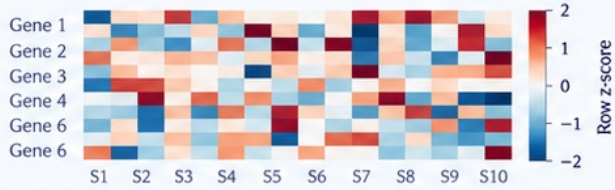
1 Raw points + 95% CI

Show individual observations and a defined interval (e.g., 95% CI) to convey uncertainty.



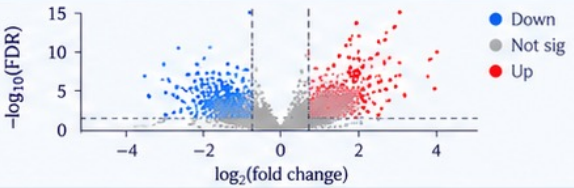
2 Zero-centred diverging palette (interpretable)

Use a perceptually uniform diverging palette with a meaningful center (e.g., 0 or baseline) and a labeled scale.



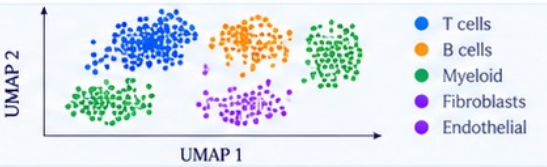
3 Full feature display (all points + thresholds)

Show all features, color by significance and direction, and indicate threshold lines.



4 Annotated embedding (with context)

Label clusters, add legend and metadata (e.g., cell type or condition), and show interpretable axes.



Clarity without scientific fidelity is decoration, not evidence.

Evidence First, Aesthetics Second

A clear workflow from question to a scientifically valid figure

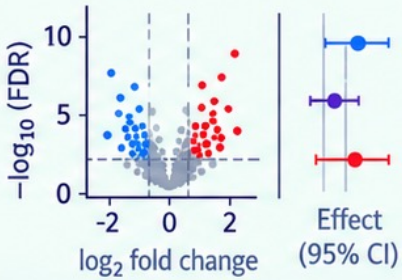
01 QUESTION
What is tested?



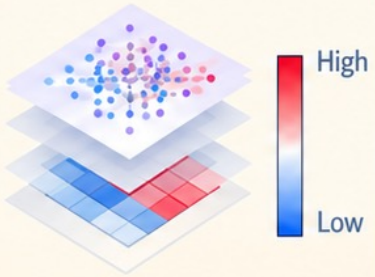
02 DESIGN
Donor / sample / cell



03 STATISTICS
Effect · FDR · CI



04 GRAPHIC
Scale · colour · layers



05 AUDIT
Check claim & source



STOP if sample IDs, model, thresholds, uncertainty, or figure provenance are missing.

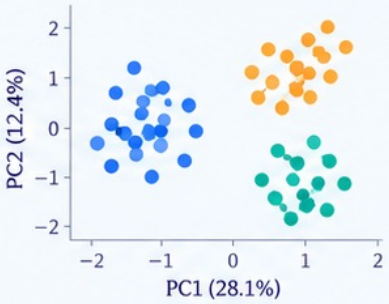
Good figures preserve the biological question, the statistical object, and the evidence.

What Does One Data Point Represent?

Different plots encode different biological units.

1 PCA / PCoA

Each point = independent biological sample



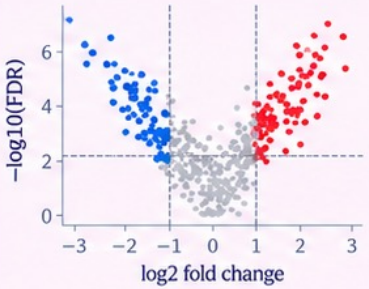
One point represents one independent sample

e.g., a patient, individual, or experimental replicate.

! **Not a hypothesis test or proof of groups.**

2 Volcano plot

Each point = one tested gene or feature



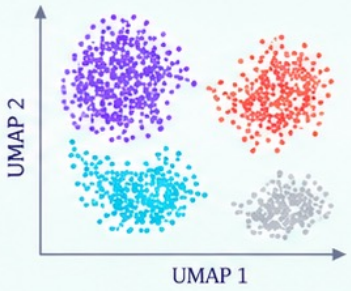
Gene / feature

One point represents one gene or feature tested for association with a condition or group.

! **Not direct causation or function.**

3 UMAP / t-SNE

Each point = cell neighborhood in an embedding

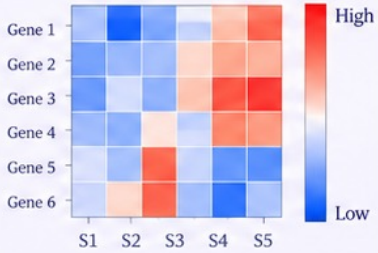


One point represents a cell (or a neighborhood of similar cells) in the low-dimensional embedding.

! **Not a physical distance or lineage map.**

4 Heatmap

Each tile = transformed value for one feature × sample

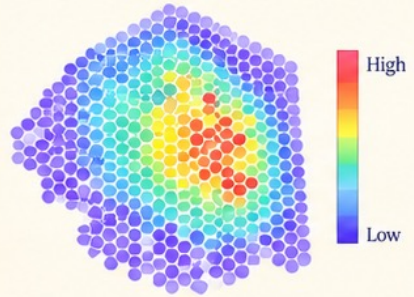


One tile represents a transformed value (e.g., normalized expression) for one gene (row) in one sample (column).

! **Not a raw count or direct measurement.**

5 Spatial plot

Each point = measured coordinate at assay resolution



One point represents a spatial spot at the assay resolution (containing multiple cells), measured at a specific tissue coordinate.

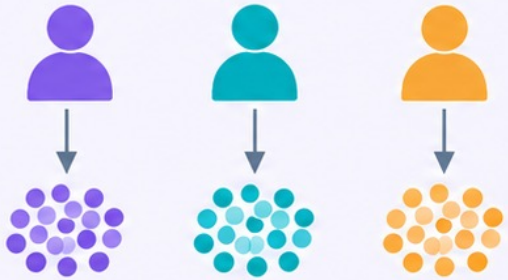
! **Not a single cell or precise location.**

Cells Are Not Independent Donors

Cell-level display and donor-level inference answer different questions

1 3 donors

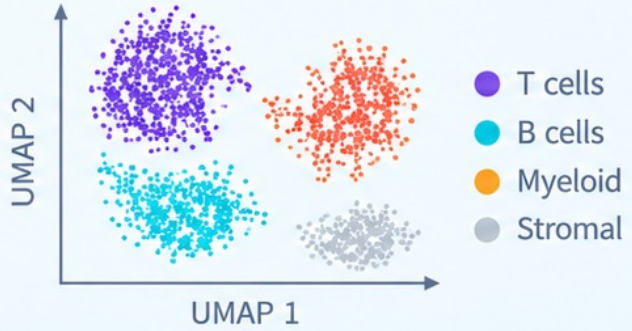
biological n = 3



many cells per donor

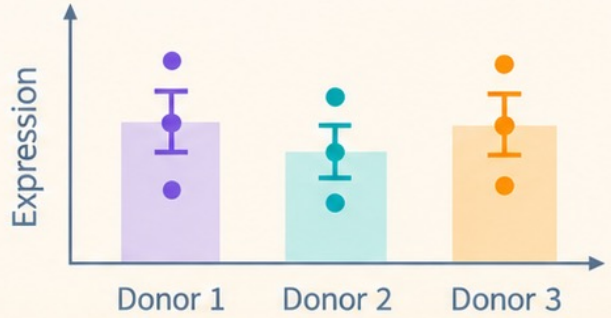
2 Cell-level display

UMAP / marker atlas



3 Donor-level inference

pseudobulk / mixed model



Cells help localization; donors establish independent replication.

- Cells provide detailed structure and cell-type characterization.
- Independent replication requires multiple biological donors.

Key terms

Pseudobulk	Aggregate counts per donor × cell type, then analyze with standard bulk methods.
Mixed model	Model donor as a random effect to account for within-donor correlation between cells.
Biological n	The number of independent donors, not the number of cells.

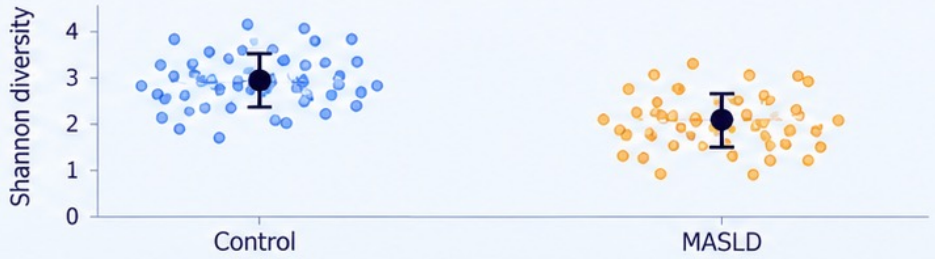
Replication depends on study design, not on the apparent number of dots.

Effect Size, Uncertainty & Statistical Evidence

Magnitude and uncertainty should be visible together

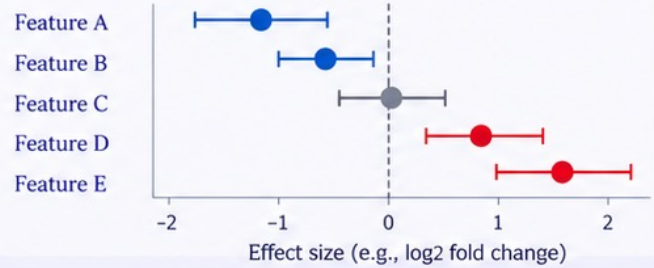
1 Raw Points + Mean + 95% CI

Show the distribution, not just a single number.



2 Effect Size (Forest Plot)

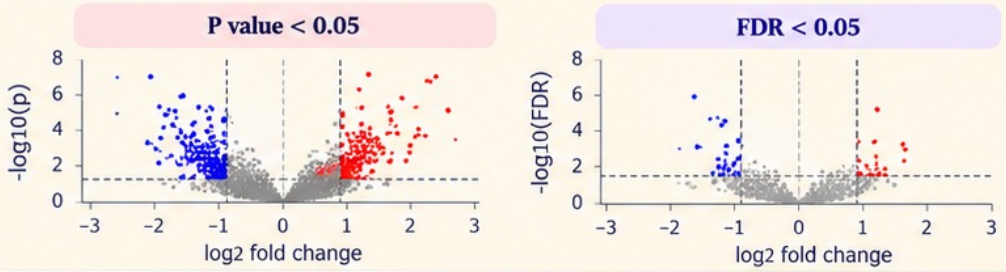
Point estimates with 95% confidence intervals.



Effect	95% CI
-1.2	[-1.8, -0.6]
-0.6	[-1.0, -0.1]
0.1	[-0.4, 0.5]
0.8	[0.3, 1.4]
1.5	[0.9, 2.1]

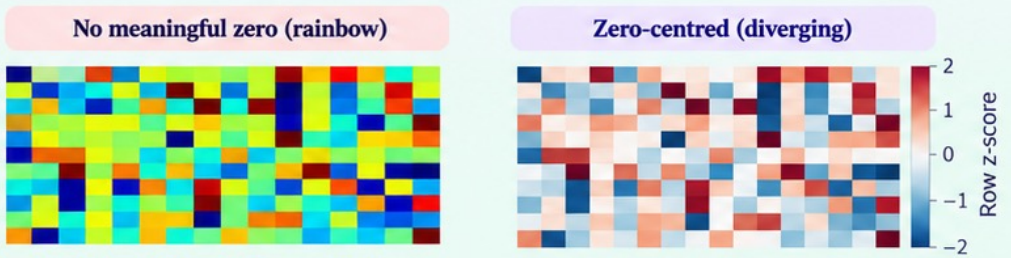
3 P value vs FDR

Many small p values may not survive multiple testing.



4 Why a Meaningful Zero Matters

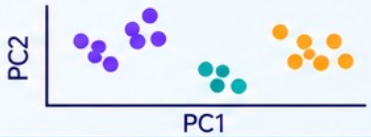
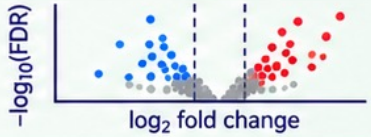
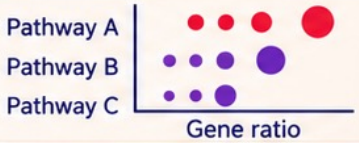
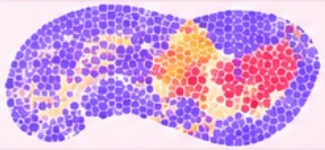
Diverging scales show direction and magnitude of change.



Report both effect size and uncertainty; significance alone is not biological meaning.

Choose the Right Plot for the Question

The analytical task determines the visual task

Difference		raw points + interval	Replicate / covariate?
Global structure		PCA / PCoA	Distance / batch?
Molecular feature		volcano / effect plot	All tests + FDR?
Pathway		GSEA / ORA dot	NES / FDR / genes?
Tissue location		spatial overlay	Scale / geometry?

The scientific question determines the plot—not personal aesthetic preference.

Explore, Diagnose or Explain?

Different figures serve different scientific purposes

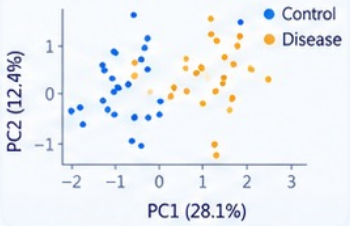
1 Exploratory

Look for patterns and generate hypotheses

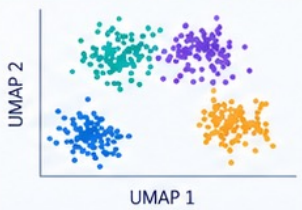
Purpose:
Explore the data, find structure, and suggest potential signals.

Example figures

Sample PCA



scRNA-seq UMAP



Helps reveal:
- Overall structure and heterogeneity
- Clusters and potential subgroups
- Outliers and batch effects

Cannot prove:
- Statistical significance
- Causal relationships

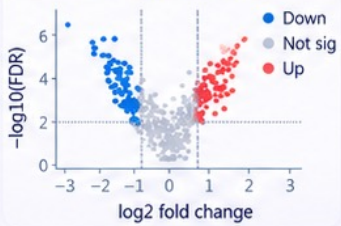
2 Diagnostic

Test specific differences and assess significance

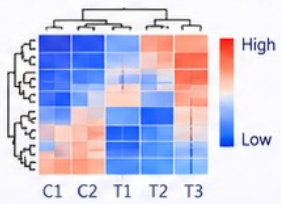
Purpose:
Identify and quantify differences between groups.

Example figures

RNA-seq volcano plot



Gene expression heatmap



Helps reveal:
- Differential features (genes, metabolites, etc.)
- Direction and magnitude of change
- Statistical significance (e.g., FDR)

Cannot prove:
- Biological mechanism
- Broader system-level impact

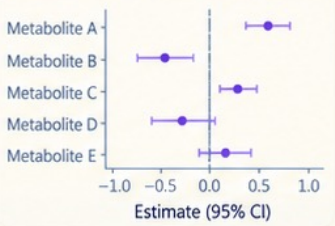
3 Explanatory

Summarize and interpret biological meaning

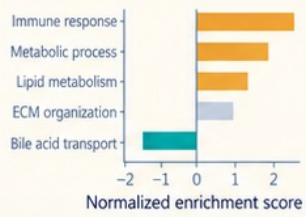
Purpose:
Translate results into biological interpretation and context.

Example figures

Metabolite effect + CI



GSEA: pathway enrichment



Helps reveal:
- Estimated effects and uncertainty
- Enriched pathways and biological processes
- Connection to prior knowledge

Cannot prove:
- Causality (association only)
- That the full biology is captured (incomplete view)

Different figure types answer different questions and require different interpretation rules.

Think Like a Scientist Before You Plot

Good prompts begin with evidence, not with aesthetics.

Think Before Prompting

1 Biological question



- Define hypothesis or goal
- Specify system and context
- Clarify comparison groups

2 Observational unit



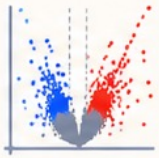
- What is one row in the data?
- Sample, cell, spot, gene, ...
- Independent replicates

3 Data table & metadata



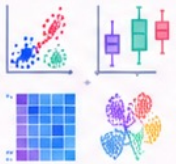
- Input data matrix
- Sample IDs and annotations
- Quality and missingness checks

4 Statistical object



- Expression, logFC, proportion, ...
- Which model or test applies?
- Assumptions and confounders

5 Plot choice



- Figure type matches question
- Labels, colors, groups, layout
- Clear and minimal design

6 Maximum defensible claim



- State what the data support
- Avoid overclaiming
- Link to evidence in the figure

AI-Assisted Workflow

01	02	03	04	05
HUMAN	AGENT	R	CRITIC	HUMAN
Question & design	Inspect data & plan	Execute analysis	Check assumptions	Approve & interpret

Reproducible Artifacts

01	02	03	04	05
DATA	SCRIPT	FIGURES	METHOD	ENVIRONMENT
input.csv + metadata	analysis.R + tests.R	PNG / PDF / SVG	captions + decisions	sessionInfo / renv.lock
Sample IDs soucre files	Versioned executable code	Publication & editing files	Units, models, thresholds	Packages & reproducibility

A strong figure plan starts from the question, the data unit, and the strongest defensible conclusion.

From a Data Frame to a Scientific Figure

Data structure → verified mapping → ggplot2 → interpretable evidence

INSPECT THE TABLE

sample_metadata.csv

sample_id	condition	Shannon	batch
S01	Control	3.15	B1
S02	Control	2.68	B1
S03	Control	3.68	B1
S04	Control	3.73	B1
S05	Control	3.33	B1
S06	Control	3.59	B1

48 simulated independent samples

MAP IN R

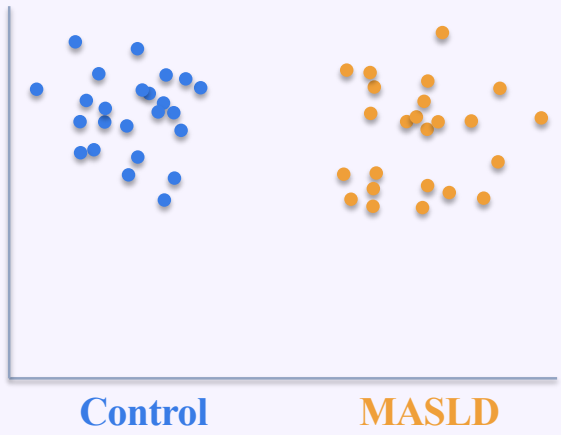
```
df <- read.csv(
"sample_metadata.csv")
ggplot(df, aes(
condition, shannon)) +
  geom_point() +

  theme_classic()
```

Check IDs | Map variables | Preserve rows

Code is editable and auditable

SHOW THE EVIDENCE



Raw points; no test implied

✓ A figure is a traceable transformation of a well-defined statistical object.

The Grammar of Graphics in ggplot2

Every visible mark has a data source, an aesthetic mapping, and a graphical rule

DATA

one row = one sample

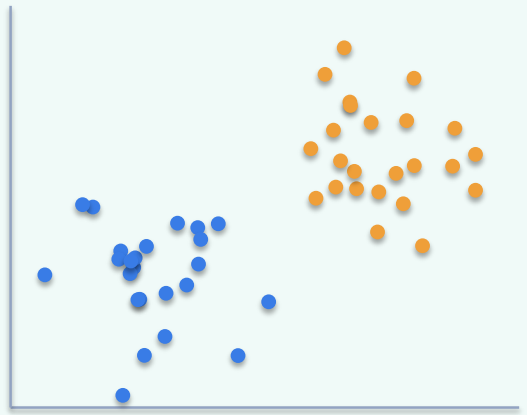
sample	group	value
S01	Ctrl	3.1
S02	Ctrl	2.7
S03	Ctrl	3.7
S04	Ctrl	3.7
S05	Ctrl	3.3

A data frame does not select its own plot.

LAYERED SPECIFICATION

1. **data = df** — observation table
2. **aes(x, y, colour)** — variable mapping
3. **geom_point()** — point geometry
4. **stat_summary()** — visual summary
5. **scale_colour_*()** — colour semantics
6. **coord_cartesian()** — axis coordinates
7. **theme_classic()** — typography / spacing

FINAL ggplot OBJECT



Aesthetic mappings

Control MASLD

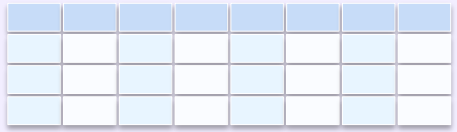
✓ **Audit the mapping and statistics before admiring the final appearance.**

Build a Figure with R — Layer by Layer

One dataset; six controlled changes to the graphical representation

01 Read data

`read.csv()`



02 Map axes

`aes(x, y)`



03 Add points

`geom_point()`



04 Encode groups

`colour = group`



05 Add summary

`geom_errorbar()`



06 Polish layout

`theme_classic()`



Plot layers change presentation; they do not grant permission to change source statistics.

Scientific Colors, Scales & Transformations

Colour encodes the variable; the transform changes the meaning of the numbers

CATEGORICAL

Different groups



Identity, not magnitude

SEQUENTIAL

Ordered magnitude



Low → high intensity

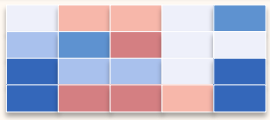
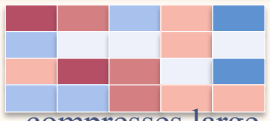
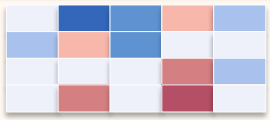
DIVERGING

Signed effects



Negative ← zero → positive

BEFORE → AFTER: NORMALIZATION

Raw counts	log2(x + 1)	Z-score
		
library size differs	compresses large values	row-centred display

AXES & DATA SEMANTICS

$\log_2$ FC	CLR	Z-score
signed effect	composition-aware	relative within feature

Show the transform, midpoint, range, and units in the figure—not only the legend.

Four Repairs That Preserve Scientific Meaning

Improve readability without changing the estimate, tested denominator, or data unit

MEANS ONLY

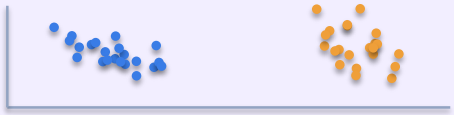
Hides the spread



→

RAW POINTS + 95% CI

Shows variation and CI



RAINBOW SCALE

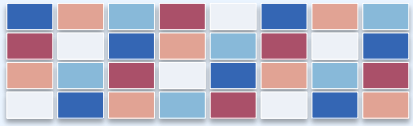
Colours lack ordered meaning



→

DIVERGING SCALE

Zero is explicitly encoded



FILTERED VOLCANO

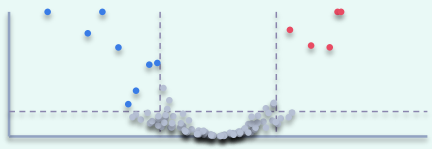
Hides tested features



→

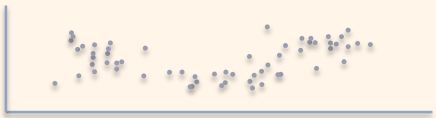
COMPLETE VOLCANO

Preserves the denominator



UNLABELED UMAP

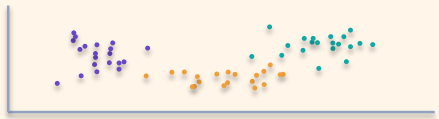
Clusters lack context



→

ANNOTATED UMAP

Displays cautious labels

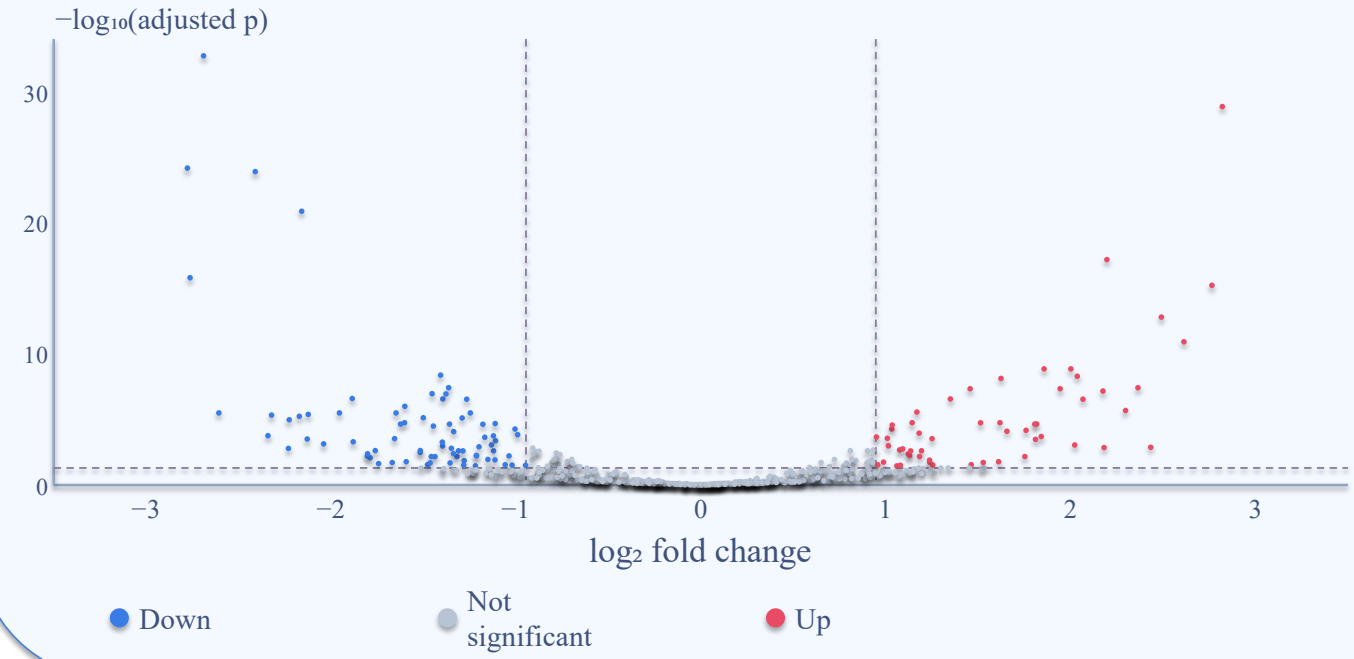


✓ **Scientific polishing should expose the data and assumptions—not alter the result.**

RNA-Seq: Volcano Plots That Tell the Truth

Effect size and multiple-testing-adjusted evidence must be shown together

700 TESTED SYNTHETIC FEATURES



READ THE PLOT

X axis
Signed $\log_2$FC

Y axis
$-\log_{10}$(padj)

Classroom rule
padj < 0.05

Effect filter
$|\log_2 FC| > 1$

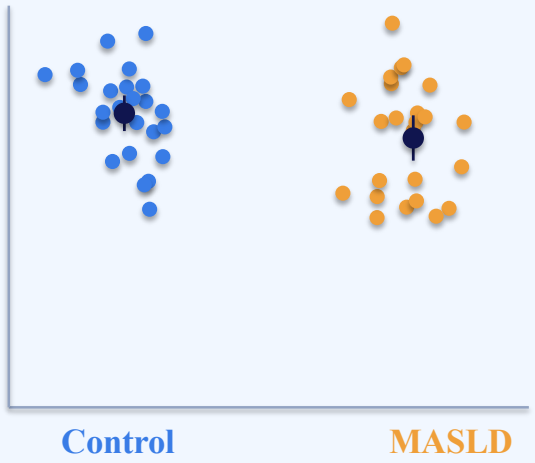
SIMULATED — NOT CLINICAL EVIDENCE

A volcano plot screens associations; it does not establish a causal gene mechanism.

Microbiome: Diversity, Distance & Composition

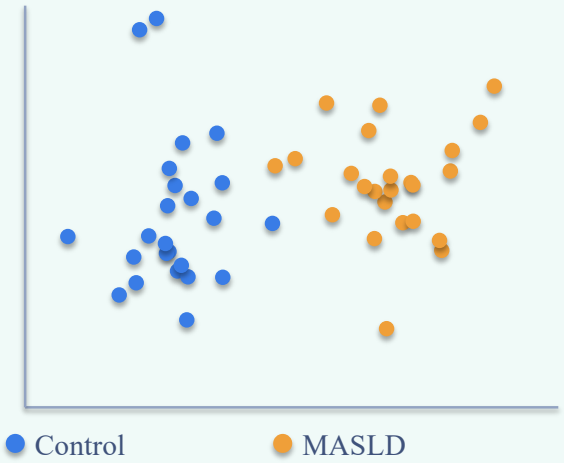
Different microbial summaries require different interpretations

ALPHA DIVERSITY



Shannon index • per sample

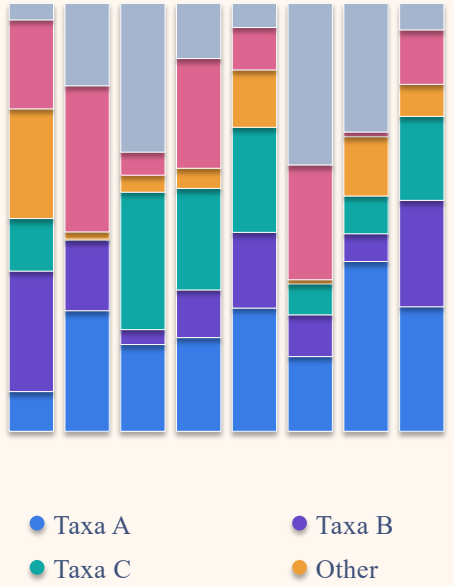
BETA STRUCTURE



Schematic ordination • no PERMANOVA

TAXONOMIC COMPOSITION

Illustrative proportions; sum = 100%



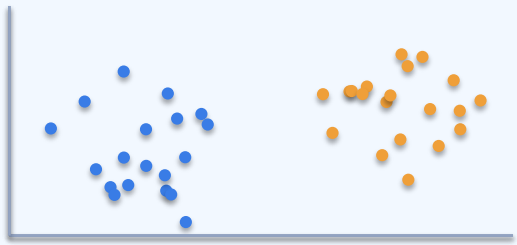
Alpha, beta, and relative abundance plots are complementary—not interchangeable.

Proteomics & Metabolomics: Patterns and Effects

QC first, then feature-level patterns and model-supported estimates

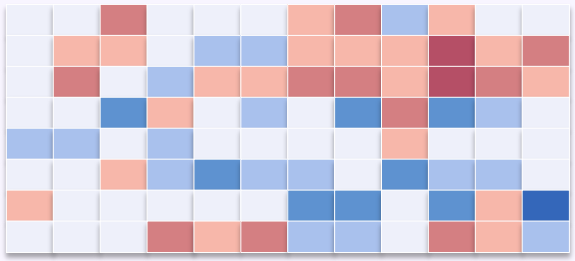
BIOSPECIMEN → LC–MS SIGNAL → FEATURE TABLE → INFERENCE + CI

QC & BATCH STRUCTURE



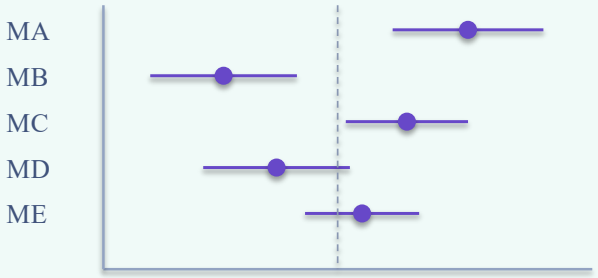
Illustrative QC pattern

NORMALIZED FEATURE HEATMAP



Feature × sample • row z-score

ESTIMATED METABOLITE EFFECTS



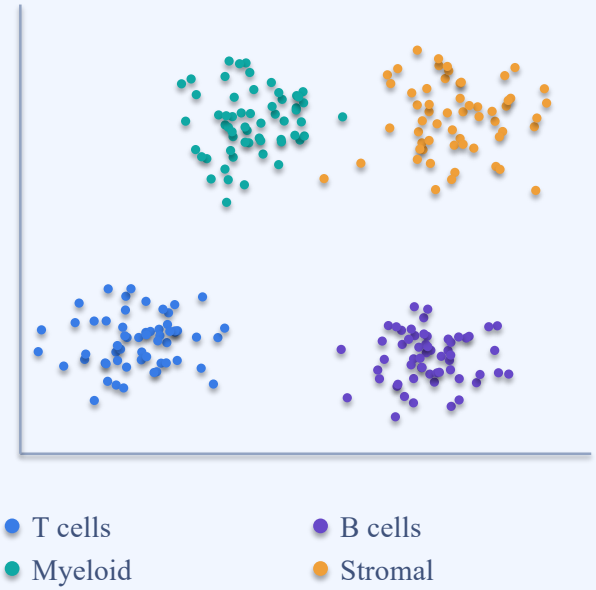
95% CI • simulated effect estimates

A molecular peak becomes interpretable only after QC, identification, and uncertainty checks.

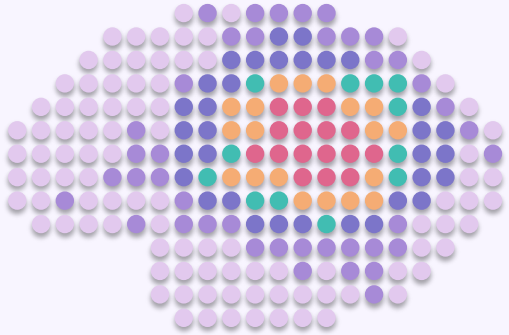
Single-Cell & Spatial Omics Visualization

Cell identity, donor replication, and tissue geometry are separate evidence layers

CELL-LEVEL ATLAS

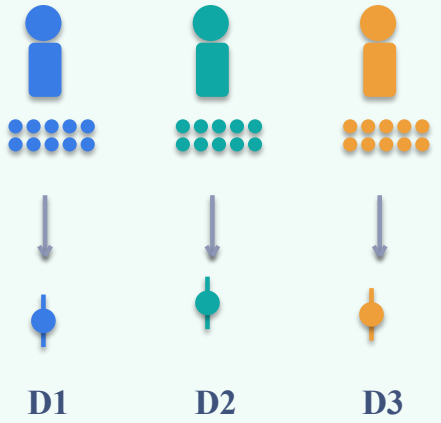


SCHEMATIC SPOT MAP



Illustrative spot-scale map

DONOR INFERENCE



Pseudobulk / mixed model

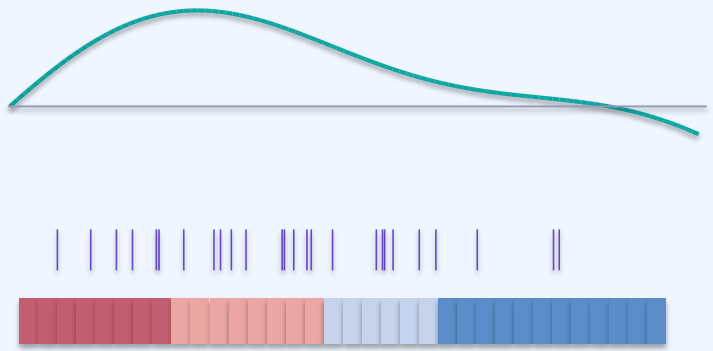
Cell-level maps locate signals; biological replication supports group-level inference.

Functional Enrichment: From Genes to Pathways

Gene-set enrichment provides statistical context—not direct evidence of pathway activity

GSEA: RANKED-LIST LOGIC

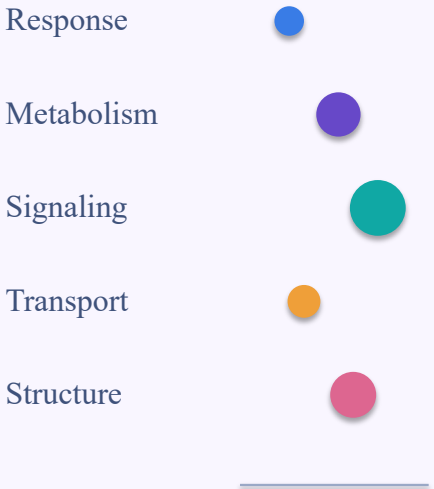
Schematic illustration • not computed from the mock CSV



Ranked genes → positive / negative enrichment

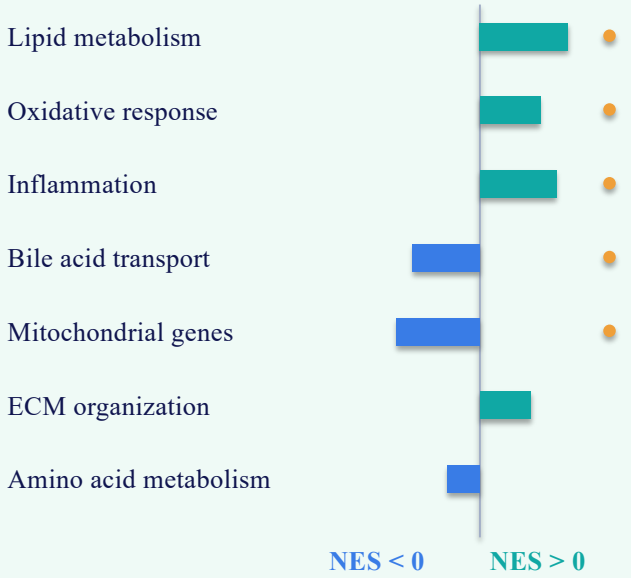
ORA DOT PLOT

Schematic gene-set ratios



Gene ratio + FDR

SIGNED NES + FDR



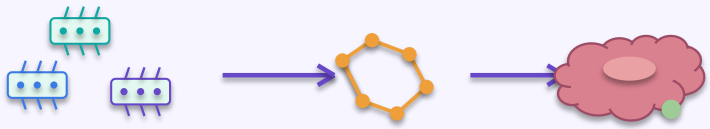
Report the ranked statistic, background, effect direction, and FDR before making claims.

The MASLD Multi-Omics Figure Challenge

One biological question • four simulated data tables • one audited figure package

ILLUSTRATIVE GUT–LIVER CONTEXT

Gut microbiota



plasma metabolites

liver molecular signal

Schematic hypothesis only

No microbiome production or causal flux was measured.

SIMULATED INPUTS & VISUAL TARGETS

sample_metadata.csv

48 records

sample · group · batch · Shannon · PC1/2

liver_DE_simulated.csv

700 records

gene · log2FC · padj · baseMean

gsea_simulated.csv

7 records

pathway · NES · padj · set size

metabolite_effects_simulated.csv

5 records

feature · effect · 95% CI · padj

The goal is a defensible figure package—not an invented gut–liver mechanism.

Understand the Data Before Asking AI

Inspect columns, independent units, missingness, and joins before making any plot

DATA MANIFEST & SEMANTIC CONTRACT

FILE	N	UNIT	KEY FIELD
sample_metadata.csv	48	independent sample	sample_id
liver_DE_simulated.csv	700	tested feature	gene_id
gsea_simulated.csv	7	gene set	pathway
metabolite_effects_simulated.csv	5	metabolite effect	metabolite

DO NOT assume feature tables are 1:1 sample-level joins.

4 QUESTIONS AI MUST ANSWER

1. **IDENTITY**
   Are IDs unique and valid?
2. **UNIT**
   Does each row mean sample, gene or set?
3. **MODEL**
   Who computed effects, FDR and CI?
4. **LINKAGE**
   Which assays actually share samples?

✓ **If a schema or independent unit is unknown, AI must ask—not infer.**

AI Copilot vs. AI Agent

Generating a code snippet is different from operating a bounded analytical workflow

AI COPILOT: ASSISTED CODING

Question
"Why is my ggplot blank?"

Code suggestion
Propose a short function

Human execution
Scientist runs & checks it

One proposal → one user-triggered step

AI AGENT: CONTROLLED EXECUTION

1. **Plan**
2. **Inspect**
3. **Code**
4. **Run**
5. **Audit**

Failure → logged repair → rerun

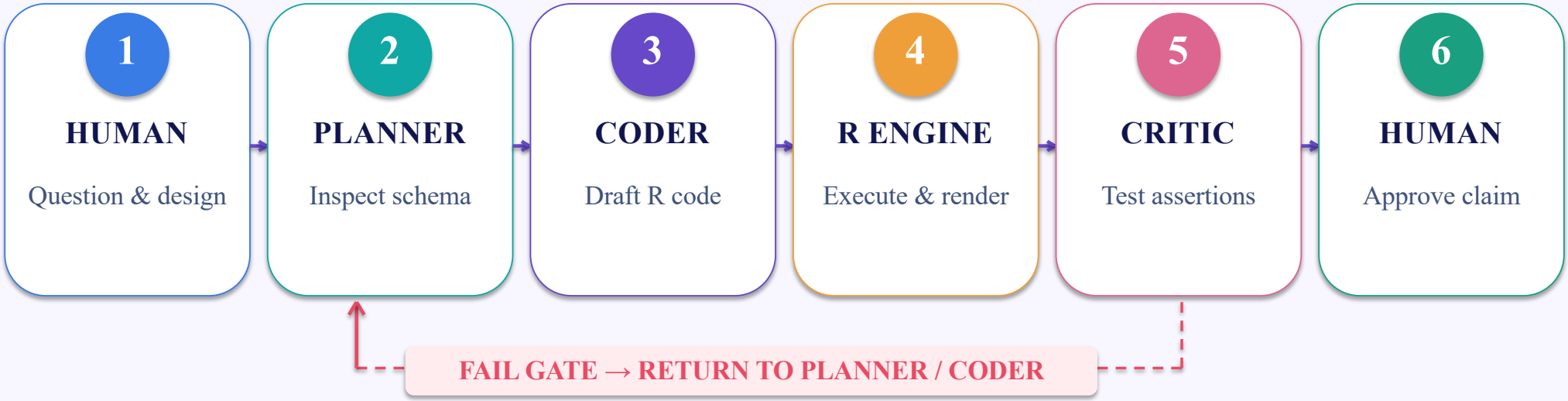
READ	Inspect schema
EXECUTE	Run approved R
APPROVE	Human decision

✓ **Agent autonomy expands execution—but never replaces human scientific authority.**

The Human–Agent–R–Critic Workflow

Human-defined scientific scope → local code execution → evidence-backed verification

SIX STAGES WITH EXPLICIT RESPONSIBILITY



Every PASS/FAIL result must reference a source field, an R assertion, or an inspected figure property.

✓ **The agent proposes; the script executes; the critic checks; the scientist decides.**

Why Vague Prompts Produce Unreliable Figures

Visual attractiveness is not a substitute for specifying data semantics and inference limits

VAGUE REQUEST

"Make a beautiful, convincing MASLD volcano."

Unknown denominator	No all-tested-feature rule
Opaque threshold	May optimize to look dramatic
Unsafe label rule	May invent disease-driving genes

A prettier plot can be a worse representation.

VERIFIABLE TASK CONTRACT

INPUT	700 genes; unique gene_id
FIELDS	log2FoldChange + padj
RULE	FDR < 0.05 and |log2FC| > 1
KEEP	ALL tested features visible
OUTPUT	R script + PNG + audit log

STOP when FDR, IDs, or contrast cannot be verified.

A strong prompt specifies scientific invariants, not an attractive outcome.

Build a Machine-Checkable AI Task Contract

Define inputs, plotting rules, prohibited actions, stop criteria, and exact outputs

TASK: VERIFIED VOLCANO PLOT

INPUT	liver_DE_simulated.csv
UNIT	One row = one tested gene
REQUIRED	gene_id · log2FoldChange · padj
CHECK	Unique IDs · finite values · $0 < padj \leq 1$
PLOT	x = log2FC; y = −log10(padj)
RULE	FDR < 0.05; |log2FC| > 1
PRESERVE	Keep 700 tested feature records

BOUNDARIES & ARTIFACTS

OUTPUT script.R · figure.png · counts · caption

NEVER Change p values or remove unselected genes

STOP Absent FDR, duplicate IDs, unknown contrast

Scientific controls are not styling options.

✓ **A task contract lets an AI agent fail safely and makes the result auditable.**

Let AI Inspect the Data — Before Plotting

Inspect names, types, independent units, value ranges, and row preservation

48	samples
700	genes
7	gene sets
5	metabolites

AUDIT-FIRST R CODE

schema_checks.R

```
d <- read.csv("liver_DE_simulated.csv")
required <- c("gene_id",
  "log2FoldChange", "padj")
stopifnot(all(required %in% names(d)))
stopifnot(!anyDuplicated(d$gene_id))
stopifnot(all(d$padj > 0 & d$padj <= 1))
n_tested <- nrow(d)  # preserve 700
```

SCHEMA GATES

- ✓ gene_id column exists
- ✓ 700 gene records retained
- ✓ No duplicate gene IDs
- ✓ padj within (0, 1]
- ! Model provenance required

Model choice is not verified by a CSV schema.

✓ Run deterministic R assertions before accepting any AI-generated chart.

Generate, Execute & Debug R Code

The agent writes code; R exposes failures; an approved repair changes the implementation

1 GENERATE R SCRIPT

plot.R

```
d <-
read.csv("liver_DE_simulated.csv"
)
ggplot(d, aes(
  log2FoldChange,
  -log10(padj))) +
  geom_point()
```

Agent: propose code

2 EXECUTE & TEST

Rscript plot.R

Error: padj not found

- Do not substitute raw p values
- Check schema and source
- Record failure and changed code

3 REPAIR & VERIFY

- Inspect names(d)
- Restore approved source schema
- Rerun assertions
- Render only when checks pass

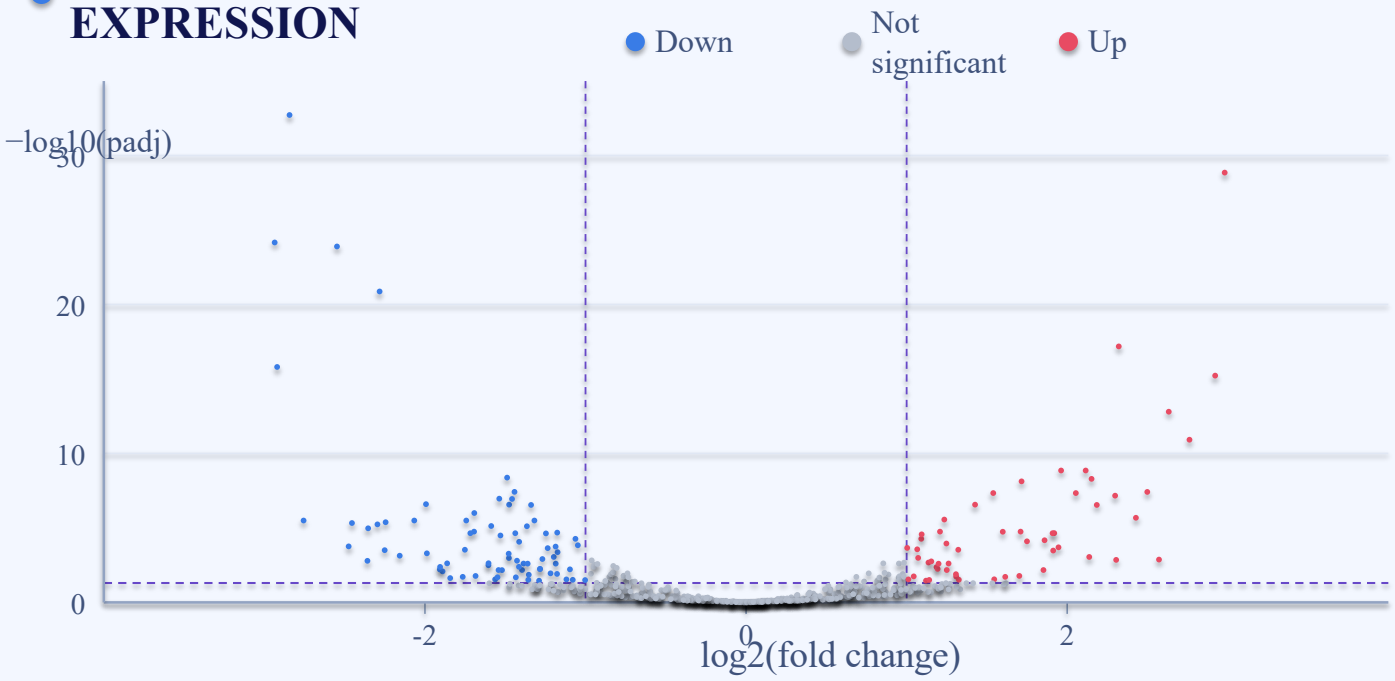
PASS → PNG / PDF / log

✓ **Debug the pipeline and preserve the evidence—never manufacture a missing field.**

Live Example: AI-Generated Volcano Plot

All 700 simulated tested genes • signed log2 fold change • supplied adjusted p values

FULL FEATURE SET — SIMULATED DIFFERENTIAL EXPRESSION



VERIFICATION

- All 700 retained
- FDR < 0.05
- |log2FC| > 1

54 Up

76 Down

Not a biological finding

The displayed categories follow fixed rules, not AI-selected thresholds.

When AI Must Stop Instead of Guessing

Failing safely is a scientific success when the requested evidence is incomplete

MISSING FDR

Only pvalue exists

Request adjusted p or upstream model output

DUPLICATE IDS

gene_id is not unique

Stop and inspect source and join rules

SILENT FILTER

AI drops nonsignificant genes

Retain full denominator or justify subset

FALSE TEST

UMAP groups look separated

Require donor-level test and assumptions

CAUSAL ARROW

Correlation presented as causation

Label hypothesis or request perturbation

PRIVACY RISK

Patient data requested by external AI

Require institutional approval

An honest STOP preserves the data, records the problem, and requests authorization.

Five Verification Gates for AI-Generated Figures

A figure becomes evidence only after its inputs, statistics, rendering, and claims survive audit

01 SCHEMA

IDs • columns • types



02 DATA

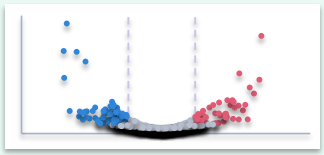
n • NA • sample unit

48 samples

700 genes

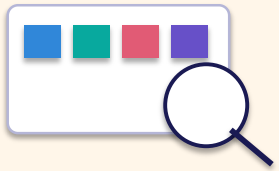
03 STATISTICS

effect • FDR • CI



04 RENDER

axes • labels • crop



05 CLAIM

evidence • limitation

PASS
PASS
REVIEW

CHECK RESULT

PASS
Matches source and contract

FAIL
Stops unsafe plotting

NEEDS INFORMATION
Requests a missing design field

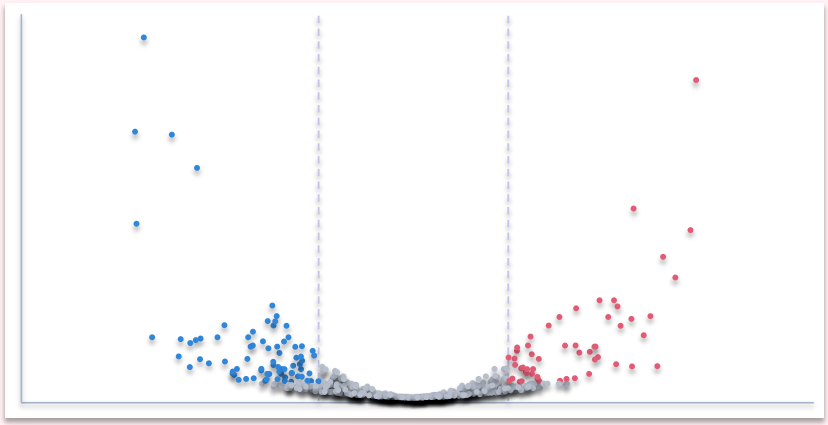
Machine-checkable R assertions + visual review + human approval

✓ **A persuasive image is not verification: every gate must reference an inspectable result.**

Visual Inspection Is Not Statistical Verification

The same attractive chart can be scientifically wrong when its labels or statistical fields are incorrect

LOOKS CONVINCING



Caption says: "FDR < 0.05"

UNVERIFIED FDR CLAIM

VERIFY AGAINST THE SOURCE

Data object / R check

names(result) • unique IDs • padj

700 Total tested features retained

130 Features meeting both display thresholds

0 Permitted silent substitutions

Require exact column mapping and a traceable model

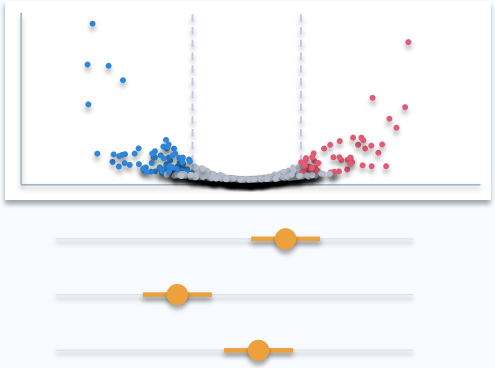
✓ **A verified caption must match the exact effect, p-value adjustment, and modeled contrast.**

From Omics Figures to Biological Interpretation

Each omics layer contributes evidence—but association and mechanism are different levels of inference

1. **RNA-seq**
   Differential expression
2. **Metabolomics**
   Effect estimates + CI
3. **Microbiome**
   Composition / diversity
4. **Cell / space**
   Localization context

ALIGN EVIDENCE



Sample pairing: REQUIRED

EVIDENCE LEVELS

- **MEASURED**
  Assay-specific observation
- **MODELED**
  Effect + uncertainty
- **INTEGRATED**
  Convergent association
- **PROPOSED**
  Testable mechanism

✓ **Cross-omics agreement prioritizes hypotheses; experimental perturbation tests mechanisms.**

Correlation Does Not Establish a Mechanism

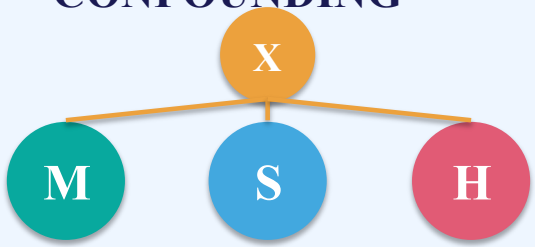
Competing causal explanations can generate similar microbiome–metabolite–liver correlations

A MICROBIAL CONTRIBUTION



Microbe → metabolite → host

B SHARED CONFOUNDING



Severity affects all three layers

C REVERSE / FEEDBACK



Host state changes microbiota

What separates the hypotheses? Tracing • perturbation • longitudinal sampling • confounder control

M = microbiome; S = circulating signal; H = hepatic phenotype; X = confounder. All diagrams are hypothetical.

Multi-Omics Concordance: What Can We Infer?

Align sample identity, assay units, and uncertainty before connecting signals across molecular layers

MATCHED DATA REQUIREMENTS

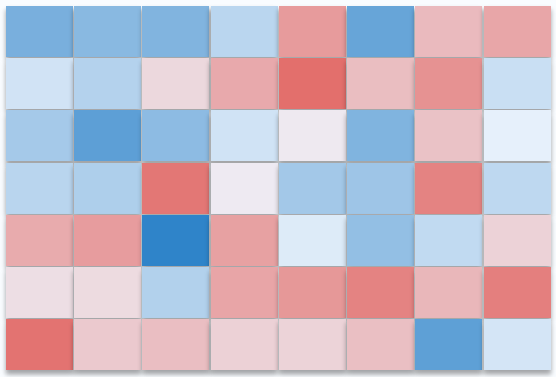
RNA profiles
S01 S02 S03 …

Metabolite profiles
S01 S02 S03 …

Microbial profiles
S01 S02 S03 …

Common sample IDs + paired n

CROSS-LAYER EVIDENCE MAP



Illustrative effect concordance

Not computed from these CSVs

INTERPRETATION LIMITS

MEASURED
Observed assay output

ASSOCIATED
Estimated statistical relation

PREDICTED
Inferred functional potential

CAUSAL
Requires perturbation

✓ **The supplied teaching dataset lacks matched per-sample multi-omics matrices: no joint correlation is computed.**

Build a Multi-Panel Scientific Storyboard

Lead readers from study quality to results, functional context, and a constrained hypothesis

A COHORT / QC

Who was measured?



n · batch · QC

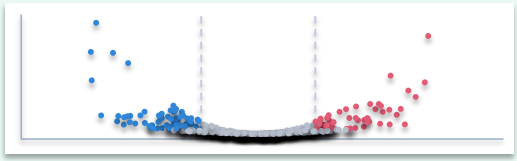
B GLOBAL PATTERN

How are samples related?



C RNA EFFECT

Which genes differ?



D PATHWAY CONTEXT

Which gene sets enrich?



E CELL / SPACE

Where is the signal?

DATA NOT PROVIDED

Use explicit placeholders

F CROSS-OMICS

What remains a hypothesis?



HYPOTHESIS ONLY

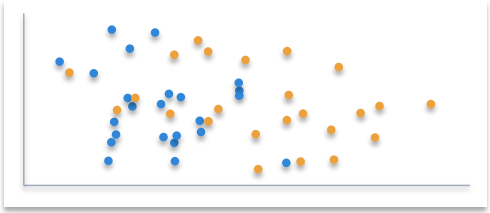
✓ **One question per panel. Exploratory plots, model-derived effects, and hypotheses must be clearly distinguished.**

Assemble Figures with patchwork

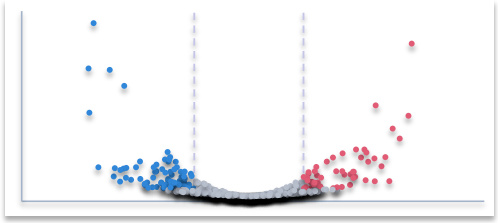
Use validated plot objects to communicate one story—composition never substitutes for statistical analysis

FOUR VALIDATED PLOT OBJECTS

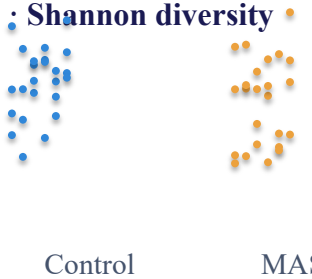
A · Sample coordinates



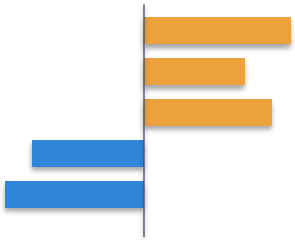
B · RNA differential expression



C · Shannon diversity



D · GSEA summaries



EDITABLE R COMPOSITION

```
library(patchwork)
figure <- (p_pca | p_volcano) /
          (p_alpha | p_gsea)

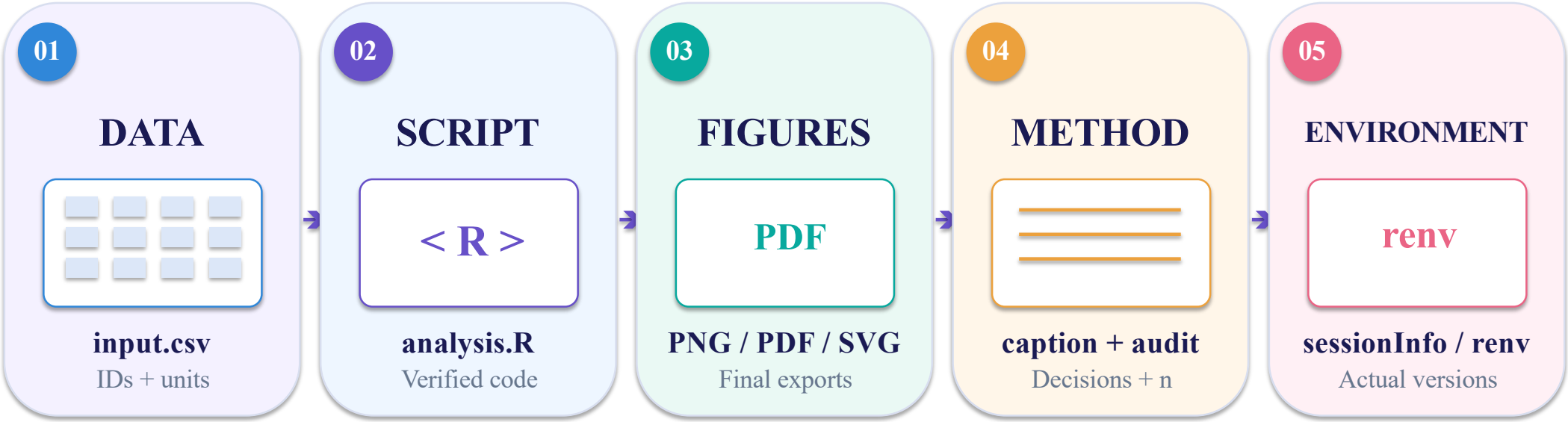
ggsave("figure.pdf", figure,
       width = 180, height = 155,
       units = "mm")
```

Layout ≠ reanalysis

Share styles and labels across compatible panels; retain each assay's own statistical meaning.

From R Output to a Reproducible Figure Package

A publication figure is an auditable collection of data, code, exports, decisions, and environments



Clean-session rerun + immutable source inputs + documented AI changes

✓ **A screenshot without source code, data provenance, and an environment record is not reproducible.**

AI as a Scientific Critic—Not Just a Plotting Tool

Ask AI to compare alternative explanations and design tests that could falsify your preferred story

H1 MICROBIAL PATHWAY

Microbial metabolism contributes to circulating signal



DECISIVE TEST

Tracer flux + microbial perturbation

H2 SHARED HOST PROCESS

Host physiology independently alters metabolites and microbiota

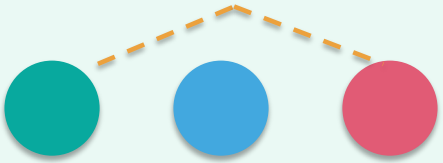


DECISIVE TEST

Time series + host pathway intervention

H3 CLINICAL CONFOUNDING

Medication or severity drives the apparent cross-omics link



DECISIVE TEST

Covariate and negative-control analysis

✓ **AI should identify missing evidence and proposed falsification experiments—not invent a causal result.**

From Beautiful Figures to Defensible Discoveries

Week 10 synthesis: scientific reasoning, transparent computation, AI assistance, and human verification

1 QUESTION
Biological hypothesis

2 DESIGN
Units + sampling

3 MODEL
Effect + uncertainty

4 VISUALIZE
R / ggplot2

5 AUDIT
Source + statistics

6 INTERPRET
Evidence + limits

7 VALIDATE
Test the mechanism

PROJECT 4
2 main figures + 1 QC panel

AI proposes • R executes • deterministic checks verify • scientists decide

✓ **The goal is not prettier omics figures—it is reproducible and biologically defensible discovery.**